LA HW

Included exercises: 1, 3, 7, 9, 11, 13, 17, 19, 23, 25, 29, 31, 35, 37, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59

Exercises 1-12

In Exercises 1–12, a matrix and a vector are given. Show that the vector is an eigenvector of the matrix and determine the corresponding eigenvalue.

Original Exercise 1

$$\begin{bmatrix} -10 & -8 \\ 24 & 18 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Original Exercise 3

$$\begin{bmatrix} -5 & -4 \\ 8 & 7 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Original Exercise 7

$$\begin{bmatrix} 4 & 6 & -5 \\ 9 & 7 & -11 \\ 8 & 8 & -11 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

Original Exercise 9

$$\begin{bmatrix} 2 & -6 & 6 \\ 1 & 9 & -6 \\ -2 & 16 & -13 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

Original Exercise 11

$$\begin{bmatrix} 5 & 6 & 12 \\ 3 & 2 & 6 \\ -3 & -3 & -7 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}$$

Exercises 13-24

In Exercises 13–24, a matrix and a scalar λ are given. Show that λ is an eigenvalue of the matrix and determine a basis for its eigenspace.

Original Exercise 13

$$\begin{bmatrix} 10 & 7 \\ -14 & -11 \end{bmatrix}, \quad \lambda = 3$$

Original Exercise 17

$$\begin{bmatrix} -2 & -5 & 2 \\ 4 & 7 & -2 \\ -3 & -3 & 5 \end{bmatrix}, \quad \lambda = 3$$

Original Exercise 19

$$\begin{bmatrix} -3 & 12 & 6 \\ -3 & 6 & 0 \\ 3 & -9 & -3 \end{bmatrix}, \quad \lambda = 0$$

Original Exercise 23

$$\begin{bmatrix} 4 & -3 & -3 \\ -3 & 4 & 3 \\ 3 & -3 & -2 \end{bmatrix}, \quad \lambda = 1$$

Exercises 25-32

In Exercises 25–32, a linear operator and a vector are given. Show that the vector is an eigenvector of the operator and determine the corresponding eigenvalue.

Original Exercise 25

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} -3x_1 - 6x_2 \\ 12x_1 + 14x_2 \end{bmatrix}, \quad \begin{bmatrix} -2 \\ 3 \end{bmatrix}$$

Original Exercise 29

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} -8x_1 + 9x_2 - 3x_3 \\ -5x_1 + 6x_2 - 3x_3 \\ -x_1 + x_2 - 2x_3 \end{bmatrix}, \quad \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

Original Exercise 31

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} 6x_1 + x_2 - 2x_3 \\ -6x_1 + x_2 + 6x_3 \\ -2x_1 - x_2 + 6x_3 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}$$

Exercises 33-40

In Exercises 33–40, a linear operator and a scalar λ are given. Show that λ is an eigenvalue of the operator and determine a basis for its eigenspace.

Original Exercise 35

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 20x_1 + 8x_2 \\ -24x_1 - 8x_2 \end{bmatrix}, \quad \lambda = 8$$

Original Exercise 37

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 - x_2 - 3x_3 \\ -3x_1 - x_2 - 9x_3 \\ x_1 + x_2 + 5x_3 \end{bmatrix}, \quad \lambda = 2$$

Exercises 41-60

In Exercises 41–60, determine whether the statements are true or false.

Original Exercise 41

ments are true or false.

If $A\mathbf{v} = \lambda\mathbf{v}$ for some vector $\mathbf{v}$, then λ is an eigenvalue of

Original Exercise 43

A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if the equation $(A - \lambda I_n)\mathbf{x} = \mathbf{0}$ has a nonzero solu-

Original Exercise 45

If λ is an eigenvalue of a linear operator, then there are infinitely many eigenvectors of the operator that corre-

Original Exercise 47

The eigenvalues of a linear operator on $\mathcal{R}^n$ are the same

Original Exercise 49

Every linear operator on $\mathcal{R}^n$ has real eigenvalues.

Original Exercise 51

Every vector in the eigenspace of a matrix A corresponding to an eigenvalue λ is an eigenvector corresponding

Original Exercise 53

The standard matrix of the linear operator on $\mathcal{R}^2$ that rotates a vector through the angle θ, where

Original Exercise 55

If $\mathbf{v}$ is an eigenvector of a matrix A, then $c\mathbf{v}$ is also an

Original Exercise 57

If A and B are $n \times n$ matrices and λ is an eigenvalue of

Original Exercise 59

If A and B are $n \times n$ matrices and λ is an eigenvalue of both A and B, then λ is an eigenvalue of AB.